

Enecogen, Netherlands

Reliable flexibility for a weather-driven energy market



The plant

Located in the Europoort industrial area of Rotterdam, the Enecogen power plant is one of the most efficient and cleanest gas-fired combined cycle power plants in Europe. Built turnkey by Siemens Energy and commissioned in 2011, the plant is a 50/50 joint venture between Eneco and EP NL. With a total installed capacity of 958 MW, Enecogen supplies electricity to approximately 1.4 million Dutch households. The plant comprises two single-shaft units, each equipped with a Siemens SGT5-4000F gas turbine and an SST5-5000 steam turbine.

Thanks to its 60% efficiency and low CO₂ emissions, Enecogen is recognized as one of Europe's cleanest and most efficient gas-fired power stations. Designed for high operational flexibility, with up to 200 starts per year, it plays a vital role in stabilizing the Dutch grid—especially during periods of low renewable generation. Due to its strategic importance and its black start capability, which allows independent power restoration within an hour, Enecogen has been officially designated as critical infrastructure by the Dutch government.

The task

As the Dutch energy market becomes increasingly weather-driven and volatile, Enecogen faced the challenge of maintaining grid stability while adapting to frequent fluctuations in supply and demand. The plant needs to operate flexibly, ramping up and down almost daily and provide black start capability to restore power in the event of a grid failure.

To prepare for future hybrid operation, Enecogen plans to integrate a battery energy storage system (BESS), enabling grid support even when the plant is offline. At the same time, it must ensure compliance with the latest cybersecurity standards and maintain high operational efficiency.

Meeting these demands requires a control infrastructure that enables seamless coordination of conventional and future assets, support advanced diagnostics functionalities, and allow for fast, reliable responses to grid signals.





Highlights

- Omnivise T3000 control system for advanced automation and diagnostics
- Flexible operation with up to 200 starts per year and minimal component wear
- Future-ready architecture enabling seamless integration of future assets such as battery storage
- Cybersecurity compliance with the latest standards and requirements thanks to Omnivise T3000
- Black start unit for autonomous grid restoration

“Reliability and flexibility of Enecogen is key. Omnivise T3000 makes it easy to integrate all key assets, which creates the best solution for reliable operation. Thanks to our black start capability, we’re ready to restore the power in the grid independently. Together, these strengths make us to a future-ready energy provider.”

Elmer Stolk, Asset Management Engineer

The solution

To meet Enecogen’s evolving requirements, Siemens Energy delivered a turnkey solution built on the Omnivise T3000 control system, which serves as the plant’s digital backbone. T3000 integrates all assets such as the gas and steam turbines, black start unit, E-houses, and future battery systems into a unified control system. This enables fully automated operation of complex processes such as black start and fast load ramping.

The system also provides real-time diagnostics, predictive maintenance, and customizable control logic to support flexible operation under dynamic market conditions. Both units are managed via a shared Omnivise Edge server (Windows Terminal Server), which was upgraded in 2023.

Designed with cybersecurity and reliability in mind, the T3000 system complies with the latest international standards and offers robust protection against both digital and physical threats. Its intuitive interface allows operators to monitor and optimize plant performance with ease, while advanced diagnostic capabilities convert raw data into actionable insights. Cybersecurity is a top priority at Enecogen. The control system architecture includes fail-safe ET 200M FS modules for both gas and steam turbines, ensuring reliable and secure operations.

Thanks to its black start capability, Enecogen can restore power independently and synchronize with the grid in less than an hour. This high level of automation and flexibility ensures the plant remains a resilient and future-ready energy provider.

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